Candidate seat No:_______________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME-AMT) Examination
March 2018

ME782 Casting and Welding Processes

Date: 29.03.2018, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. One can refer **books/notes**. Exchange of books/notes is not allowed during examination.

SECTION – I

Q - 1 Answer following multiple choice type questions. **[10]**

(i) Which of the following ferrous metals have carbon more than 2 percentage:
(p) Cast iron (q) Carbon steel
(r) Stainless steel (s) All of the above

(ii) Which of the following metals have density less than 7000 Kg/m3:
(p) Copper (q) Aluminum
(r) Iron (s) All of the above

(iii) Which of the following Indian foundry clusters is hub for Investment Casting Process:
(p) Coimbatore (q) Kolhapur
(r) Rajkot (s) Belgaum

(iv) Which of the following is filling related defects:
(p) Blow hole (q) Hot tear
(r) Shrinkage (s) All of the above

(v) During mold filling, there occurs:
(p) Oxidation of advancing front (q) Change in physical properties
(r) Stream separation and merging (s) All of the above

(vi) In metal casting process, sources of gases are:
(p) Entrapped Air (q) Diffused gases
(r) Core gases (s) All of the above

(vii) In general for sand casting, high resistance to heat transfer is offered:
(p) In solidified metal (q) At metal-mold interface
(r) At liquid-solid interface (s) In mold material

(viii) During solidification, phase change occurs at constant temperature in:
(p) Pure metals (q) Hypo-Eutectic alloys
(r) Eutectic alloys (s) Hyper-Eutectic alloys

(ix) Feeding distance depends on:
(p) Size of the feeder (q) Location of the feeder
(r) Thickness of the casting (s) All of the above

(x) Draft allowance is provided on:
(p) Faces normal to draw direction (q) Faces along the parting plane
(r) Faces parallel to draw direction (s) All of the above

Q - 2 For an aluminum alloy (10% Si) cuboid casting, having 20 X 30 X 50 cm dimensions, design a cylindrical top feeder along with the connecting neck to ensure it is free from shrinkage porosity. Then calculate the yield. Take height to diameter ratio of feeder as 2. **[09]**

OR

Q - 2 (a) For an aluminum alloy (10% Si) cuboid casting, having 20 X 30 X 50 cm dimensions, poured into a silica sand mold, compute the solidification time. Take $k_{mold} = 0.8\ W/m\text{-}K$, $\rho_{mold} = 1500\ Kg/m^3$, $C_{mold} = 1100\ J/Kg\text{-}K$, $\rho_{metal} = 2700\ kg/m^3$, $C_{metal} = 900\ J/kg\text{-}K$, $\Delta H_{metal} = 400\ KJ/Kg$, $T_{melting} = 600^oC$, and ambient temperature = $30\,^oC$. **[04]**

Q - 2 (b) For above cuboid casting, compute filling time if gate of 1 sq. cm cross-section area is connected at the top of the casting. Consider sprue height as 20cm. **[05]**

Q - 3 Attempt *any four*: **[16]**

(a) Why foundry engineers want to ensure directional solidification inside cast components? Discuss the means to achieve the same with illustration.

(b) Discuss the influence of pouring at under superheat and over superheat temperature on quality of castings.

(c) "Feeder efficiency can be improved by connecting more number of cavities with common feeder". Justify the statement with brief discussion.

(d) Discuss various elements of cost involved during cast product manufacturing.

(e) State the difference between shrinkage porosity and gas porosity with illustration.

SECTION – II

Q - 4 Answer following multiple choice type questions. **[10]**

(i) Which one of the following welding processes consist of minimum heat affected zone:
(p) Shielded metal arc welding (q) Ultrasonic welding
(r) Metal inert gas welding (s) Laser beam welding

(ii) In resistance welding, the resistance should be highest:
(p) Inside electrode (q) At interface between workpiece and electrode
(r) Inside workpiece (s) At interface between two workpieces

(iii) The electrodes used in welding are coated. This coating is not expected to:
(p) Provide protective atmosphere to weld (q) Prevent electrode from contamination
(r) Stabilize the arc (s) Add alloying elements

(iv) Which of the following belongs to the category of plastic welding?
(p) Explosion welding (q) Resistance welding
(r) Gas welding (s) Laser beam welding

(v) Arc blow occurs due to:
(p) Magnetic field around the electrode (q) Ionization of arc
(r) Electric field around the electrode (s) Ionization of surrounding air

(vi) The heat generated in resistance welding is expressed by:
(p) I^2Rt (q) IR^2t
(r) IRt^2 (s) None of the above

(vii) Which of the following welding techniques uses a consumable electrode?
(p) Laser welding (q) Thermit welding
(r) TIG welding (s) MIG welding

(viii) For welding plates of thickness less than 5 mm, its edges:
(p) Do not require beveling (q) Should have double-V or U-groove
(r) Should be beveled to single-V or U-groove (s) None of the above

(ix) In general, voltage required for heating in electric resistance welding is in the range:
(p) 1 to 5 volts (q) 6 to 10 volts
(r) 11 to 20 volts (s) 50 to 100 volts

(x) The brazing process is carried out in the temperature range:
(p) Below 450˚C and above solidus temperature of weld metal (q) Below 450˚C
(r) Above 450˚C and below solidus temperature of weld metal (s) Above 450˚C

Q-5 Differentiate between the Friction welding and Friction Stir welding, giving scientific reasoning. **[05]**

OR

Q-5 Differentiate between Design for Manufacturing and Design for Manufacturability in the context of either cast or weld components. **[05]**

Q-6 Attempt *any Five*: **[20]**

(a) Discuss the reasons for generation of residual stresses in welded joints.

(b) Explain the variation in mechanical properties across the heat affected zone (HAZ) in a butt-welded joint using arc welding process.

(c) Briefly discuss various sources of heat that can be used to perform welding operation.

(d) Why welding flux is an essential aspect of welding process such as submerged arc welding?

(e) Explain why is the tungsten inert –gas welding preferred for welding aluminum plates.

(f) “In automatic arc welding, flat power source characteristic is desirable”. Justify the statement with neat sketch.

(g) Briefly explain the working principle of the Submerged Arc Welding (SAW) process and mention their applications.

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